

Promit Das

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Education

Memorial University of Newfoundland

September 2023 - Present

Bachelor of Science in Computer Science And Statistics

Personal Projects

[Personal Portfolio Website](#) | Node.js, PowerShell, HTML5, CSS ([code](#))

- Engineered a custom Static Site Generator (SSG) using vanilla Node.js to transform JSON data and HTML templates into a static website, eliminating the need for external frameworks.
- Developed a zero-dependency HTTP server with custom routing, MIME type handling, and an in-memory caching layer to optimize content delivery speeds.
- Automated the deployment pipeline with a PowerShell script that builds artifacts and syncs them to university servers via SSH/SCP.

[Flappy Bird AI \(Neuroevolution\)](#) | JavaScript, p5.js, Genetic Algorithms, Neural Networks

- Implemented the NEAT algorithm (NeuroEvolution of Augmenting Topologies) from scratch to train a population of 1,000 agents to master gameplay mechanics without human intervention.
- Engineered a custom dynamic Neural Network library that supports topological growth, allowing agents to evolve complex architectures by adding nodes and connections over generations.
- Developed evolutionary operators including crossover, variable mutation rates (weights/topology), and speciation to protect innovative strategies from being prematurely culled
- Built a real-time visualization engine using p5.js to render the active neural network structure and gameplay, featuring time-scale acceleration and model serialization (JSON export/import).

[Neurodivergent Productivity App](#) (In Development) | Next.js, FastAPI, MongoDB, Groq API(Llama3.1), Vercel, Render

- Planning an AI-driven productivity app tailored for neurodivergent users (ADHD, Autism, Dyslexia).
- Designing task-breakdown intelligence using LLMs (Groq + Llama 3.1) to convert complex tasks into actionable micro-steps.
- Mapping a focus-mode experience with adaptive timers, dopamine-boost triggers, and distraction management workflows.
- Outlining a personalized scheduling engine informed by user habits, energy cycles, and emotional checkins.
- Architecting a scalable backend with REST APIs, authentication, and NoSQL data models using tools (MongoDB Atlas).
- Preparing for deployment using AWS (S3, Lambda, API Gateway) to demonstrate cloud-ready architecture.

[Pac-Man Browser Game](#) | JavaScript (ES6+), HTML5 Canvas, Web Audio API ([code](#))

- Engineered a custom game loop using requestAnimationFrame and delta-time calculations to render smooth 60fps animations on an HTML5 Canvas.
- Synthesized dynamic sound effects (Waka-waka, sirens) programmatically using the Web Audio API oscillators and gain nodes, eliminating the need for external audio assets.
- Implemented heuristic-based AI for ghost movement, calculating Euclidean distance to target tiles to simulate "chase" and "scatter" behaviors dynamically.
- Developed a grid-based collision system with input buffering, allowing users to queue directional moves for precise cornering.
- Built a responsive touch-control interface for mobile devices, mapping pointer events to game logic for cross-platform playability.

Experiences

Software Developer | Paradigm

Student Design Hub – Memorial University | July 2024 - Present

- Contributed to an autonomous go-kart project for the Purdue University racing competition, gaining hands-on experience in real-time robotics systems
- Worked with ROS2 (nodes, topics, publishers/subscribers) and Linux (Ubuntu) for distributed system communication.
- Explored autonomy stack components including perception, planning, mapping, and control.
- Assisted with repository structure, sensor integration concepts, and multi-node debugging.

Web Developer | MUN Quant Society (Website Pod)

Memorial University | Jan 2025 - Present

- Developed and maintained society website pages using HTML, CSS, and JavaScript.
- Improved UI/UX through layout, navigation, and responsiveness updates.
- Collaborated via GitHub workflows to manage code changes and integrate content.

CAT Math Help Centre Teaching Assistant

Memorial University of Newfoundland, St. John's, NL | May 2024 - August 2024

- Provided one-on-one math support to 20+ students weekly, simplifying complex concepts in Calculus and Algebra.

Research Assistant| Education Department (MUCEP)

Memorial University of Newfoundland, St. John's, NL | May 2024 – December 2024

- Conducted comprehensive online research, summarizing and organizing 50+ academic articles into clear, structured reports.
- Streamlined documentation processes, improving information access and workflow efficiency for the research team.

Volunteering Experience

Let's Talk Science | Volunteer

St. John's, NL | May 2024 – Present

- Led hands-on STEM activities for 40+ K–5 students, simplifying science concepts for diverse learners.
- Coordinated materials and logistics for 10+ events, ensuring smooth operation.

Secretary | Memorial Math & Stats Society

Memorial University | September 2024 – Present

- Provided timely updates to 50+ members and managed society communications.
- Documented meeting minutes and coordinated event logistics.
- Supported student engagement and helped promote society initiatives